

Smart Traffic Lights Using Local Artificial Intelligence

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Abstract—Smart traffic-light systems can use traffic information to adjust signal timing according to changing road conditions. Traditional fixed-time traffic lights are simple and inexpensive, but they may not respond effectively to sudden changes in traffic demand. This paper presents a conceptual local-AI traffic-light architecture in which street cameras provide traffic information to local computing devices for rapid processing. Selected information can then be forwarded to a cloud server for storage and wider analysis. The proposed architecture is also extended to address emergency vehicle priority, with the objective of coordinating nearby traffic signals to provide a clear route for an ambulance.

Index Terms—Smart Traffic Lights, Artificial Intelligence, Machine Vision, Traffic Signal Control, Edge Computing, Emergency Vehicles

I. INTRODUCTION

Smart traffic-light systems are being studied as a way to reduce waiting time and respond more quickly to changing road conditions. Traditional traffic lights commonly operate using predetermined timing plans. Although such systems are simple to implement, they may not respond effectively when traffic conditions change unexpectedly.

Artificial intelligence and machine vision provide methods for estimating traffic conditions from camera information. Research on deep reinforcement learning has investigated the use of learning-based methods for traffic signal control and reviewed their potential for addressing changing traffic conditions [1].

Machine-vision-based approaches have also been investigated for adaptive traffic-light control. Gorodokin et al. studied the use of machine vision and neural-network-based vehicle tracking for adaptive traffic-light control [2].

The proposed work focuses on a local-AI architecture in which traffic information is processed close to the road. This approach can reduce the amount of information that must be transmitted to a remote server and can support rapid local decisions.

II. RELATED WORK

Rasheed et al. presented a review of deep reinforcement learning for traffic signal control, covering learning architectures, traffic signal control characteristics, simulation platforms, complexity, and research challenges [1].

Gorodokin et al. investigated adaptive traffic-light control using machine vision. Their work used camera-based traffic information and a neural-network-based approach to estimate traffic-flow parameters and adapt signal operation [2].

Emergency vehicle priority is another important aspect of intelligent traffic signal control. Cao et al. proposed an emergency-vehicle-centered traffic signal control method designed to reduce emergency-vehicle delay while limiting the effect on conflicting traffic directions [3].

These studies provide a foundation for developing a system that combines traffic monitoring, local processing, adaptive signal control, and emergency vehicle priority.

III. PROPOSED SYSTEM

The proposed architecture consists of four main components:

- 1) Street Cameras
- 2) Smart Pole Computers
- 3) City Cloud Server
- 4) Traffic Lights

The basic information flow is:

Street Cameras → Smart Pole Computers → City Cloud Server → Traffic Lights

The smart pole computer performs local processing of camera information and generates a traffic-control decision. Selected information can subsequently be transmitted to the cloud server for storage and wider analysis.

IV. SYSTEM FLOWCHART

The proposed workflow is shown in Fig. 1.

V. SYSTEM ARCHITECTURE

The system architecture is based on distributed processing. Street cameras capture the current traffic conditions. The captured information is processed by smart pole computers located close to the traffic intersection.

The local processing unit can estimate traffic conditions and generate a control decision. Selected information can be sent to the city cloud server for storage and wider analysis.

Finally, the traffic controller receives the decision and changes the traffic-light signals.

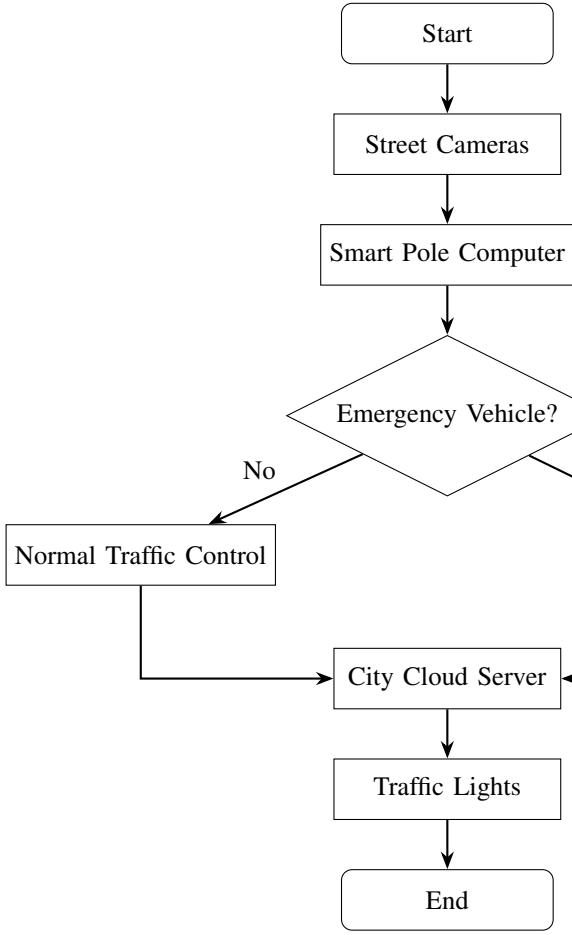


Fig. 1. Flowchart of the proposed smart traffic-light system.

VI. ARCHITECTURE DIAGRAM

The detailed architecture is shown in Fig. 2.

Place the architecture image in the images folder and name it system-architecture.png.

VII. TRAFFIC PROCESSING

The system processes traffic information in several stages. First, camera data is acquired from the road. Second, the local computer processes the camera information. Third, traffic conditions are estimated and classified. Finally, a signal-control decision is generated.

Machine-vision-based traffic monitoring has been investigated for adaptive traffic-light control, including estimation of vehicle and queue characteristics [2].

VIII. EMERGENCY VEHICLE PRIORITY

An important research gap considered in this work is the ability to automatically provide a clear route for an emergency vehicle, such as an ambulance.

Emergency vehicle signal preemption has been investigated as a means of reducing delay at intersections. Cao et al. proposed an emergency-vehicle-centered traffic signal control

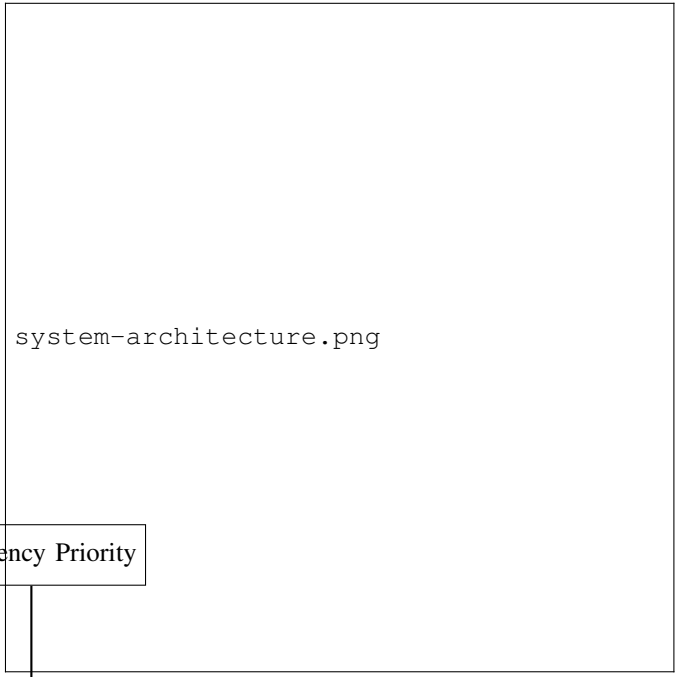


Fig. 2. Architecture of the proposed local-AI traffic-light system.

method that focuses on emergency-vehicle performance while considering the effect on conflicting traffic directions [3].

In the proposed architecture, the local computer can identify an emergency-vehicle condition and initiate an emergency-priority control sequence. Nearby signals can then be coordinated to support the emergency vehicle's movement.

IX. MATHEMATICAL MODEL

Let N represent the number of detected vehicles and let T represent the observation period.

The average vehicle arrival rate can be represented as

$$\lambda = \frac{N}{T}. \quad (1)$$

The traffic density can be represented by

$$D = \frac{N}{L}, \quad (2)$$

where L is the observed road length.

A simple traffic priority score can be defined as

$$P = w_1N + w_2Q + w_3E, \quad (3)$$

where Q represents queue length, E represents an emergency priority indicator, and w_1 , w_2 , and w_3 are weighting parameters.

X. ARRAY REPRESENTATION

The detected traffic information can be represented using a feature matrix:

Algorithm 1 Local-AI Traffic Signal Control

Require: Camera input I **Ensure:** Traffic signal decision S

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1: Capture camera frame  $I$ 
2: Detect and count vehicles
3: Estimate queue length
4: Detect emergency vehicle condition
5: if emergency vehicle is detected then
6:   Activate emergency priority
7:   Coordinate nearby traffic signals
8: else
9:   Calculate traffic priority score
10:  Select appropriate signal phase
11: end if
12: Send selected information to cloud server
13: Apply signal decision  $S$ 
    return  $S$ 
```

TABLE I
CONCEPTUAL COMPARISON OF TRAFFIC-CONTROL APPROACHES

Parameter	Basic	Cloud AI	Local AI
Decision Time	30 s	2 s	0.5 s
Internet Usage	None	High	Low
Setup Cost	\$200	\$2500	\$1200
Local Processing	No	No	Yes

$$X = \begin{bmatrix} v_1 & q_1 & s_1 & e_1 \\ v_2 & q_2 & s_2 & e_2 \\ v_3 & q_3 & s_3 & e_3 \end{bmatrix}, \quad (4)$$

where v represents vehicle count, q represents queue length, s represents estimated speed, and e represents the emergency indicator.

XI. PROPOSED ALGORITHM

The proposed processing procedure is given in Algorithm 1.

XII. SYSTEM COMPARISON

The conceptual comparison of the three traffic-control approaches is presented in Table I.

The values in Table I are the conceptual values provided in the assignment handout and should therefore not be interpreted as experimentally validated measurements. The handout gives the corresponding comparison values for basic timed lights, cloud AI systems, and local smart cameras. [1]

XIII. RESEARCH GAP

Existing traffic signal systems can operate effectively under regular traffic conditions. However, emergency situations introduce an additional requirement: the traffic-control system should be capable of rapidly identifying an emergency vehicle and coordinating nearby signals.

Research on emergency-vehicle-centered signal control demonstrates that emergency priority can be incorporated into

traffic signal control while considering the effects on other traffic [3].

Therefore, the proposed research focuses on combining local traffic monitoring with emergency vehicle detection and coordinated signal control.

XIV. DISCUSSION

Local processing provides an architecture in which traffic information can be analyzed close to its source. This can reduce the quantity of raw video information that must be transmitted to a remote server.

Machine vision provides a mechanism for extracting traffic-flow information from camera data, while learning-based traffic signal control provides methods for adapting signal decisions to traffic conditions [1], [2].

Emergency vehicle priority adds another control condition that requires rapid coordination of nearby traffic signals [3].

XV. LIMITATIONS

The proposed architecture is conceptual and requires experimental validation before practical deployment.

The actual processing time, internet usage, setup cost, and traffic-performance improvements depend on the selected hardware, software, traffic environment, camera configuration, and experimental methodology.

Therefore, the numerical comparison in this paper should not be treated as a measured experimental result unless it is supported by an actual implementation and experiment.

XVI. FUTURE SCOPE

Future work can implement the proposed architecture using a real-time traffic dataset and an embedded computing platform.

Further research can investigate:

- Real-time vehicle detection.
- Emergency vehicle recognition.
- Adaptive traffic signal timing.
- Multi-intersection coordination.
- Edge and cloud cooperation.
- Real-world traffic experiments.
- Quantitative evaluation of emergency response time.

XVII. CONCLUSION

This paper presented a conceptual smart traffic-light architecture based on local artificial intelligence.

The proposed system uses street cameras to obtain traffic information, smart pole computers to perform local processing, cloud infrastructure for storage and wider analysis, and traffic signals to implement the resulting control decisions.

The system is further extended to support emergency vehicle priority. Existing research demonstrates the relevance of both learning-based traffic signal control and machine-vision-based adaptive traffic management [1], [2].

Emergency-vehicle-centered traffic signal control has also been studied as a mechanism for reducing emergency vehicle delay [3].

Future implementation and experimental evaluation are required to determine the actual performance of the proposed system.

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